

AI Wars in 2025 — can generative AI pay for itself?

Wu, A. & Yang, A. (2025). "AI Wars in 2025." Harvard Business School Case 9-725-484, revised 25 August 2025. Prepared from published sources. Licensed to this course (PE6201, NTU) from July 2026 to January 2027.

ASSIGNED READING

Read the case — you have your own licensed copy ~50 min

If short on time: read "Google's Challenges for Search" and "Unit Economics of Generative AI in Search" in full, then skim the rest ~20 min

New to business-economics vocabulary? Read the appendix at the end of this brief first ~10 min

» ~50 min • this summary + appendix ~20 min

Where to get it. You should already have your own licensed copy of the case through the course — it is not attached to this brief. If you cannot find it, ask before class rather than after. The case is licensed to PE6201 only, so please do not forward it outside the class. This brief is a study aid, not a replacement; the case is the authority and we will work from it directly in class.

Come with the arithmetic, not the answer. We will run the numbers live in Capsule 3 — including scenarios that are not in this brief. Your job before class is to understand what each number means and what could move it, not to arrive with a verdict. If you read only one thing, read the unit-economics section; three numbers from it carry the whole session.

Note on currency. The case freezes in June 2025 — over a year before our class. Several things have moved since. Read it as a decision problem at a moment in time, not as a report on where the industry is today, and resist arguing from hindsight: "we now know what happened" is not analysis. The interesting question is what you would have done in that room, with those numbers.

KEY TAKEAWAY

Google invented the technology that is now eating its own business model, and the problem is not technical. Search earns Google an estimated 1.61 cents per query and costs about 1.060 cents to serve — a margin of roughly half a cent, multiplied by more than five trillion searches a year. Adding basic generative AI to the answer is estimated to add 0.356 cents of inference cost to every single search. That one third of a cent consumes about two-thirds of the margin. Nothing about that sentence is a research problem: Google has the models, the chips, the data and nearly \$100 billion in cash. The question is whether a product that has printed money for two decades can survive being rebuilt on a cost base that scales with every word a user types.

WHY IT MATTERS FOR THIS COURSE

This is Leg B of the Class-3 case, and it is the counterweight to Leg A. Med-PaLM 2 proved that stacking inference-time techniques — few-shot, chain of thought, self-consistency, ensemble refinement, chain of retrieval — can push a model to near-expert performance. Every one of those techniques multiplies the number of tokens spent per answer, and Google itself rationed the most expensive one. AI Wars takes that same multiplier and applies it to a product with five trillion transactions a year. The hinge between the two legs is a single question: accuracy is bought in tokens, so who pays, and can the price be passed on? Answer that and you can reason about your own build — the extraction prompt you watch us tune in Capsule 2 has exactly the same structure, four orders of magnitude smaller.

Feeds Class 3: C2 (the prompting ladder, now priced) • C3 Leg A → the hinge → Leg B — the unit economics of inference at scale. Also feeds Week 5 (strategy and economics).

MAIN POINTS

The situation — June 2025

- **The spark.** Alphabet shed a quarter-trillion dollars of market capitalisation in a matter of months. The trigger: Apple revealed in May 2025 that Google searches on Safari had fallen for the first time in over twenty years, attributed largely to users shifting to generative AI alternatives. Alphabet's share price fell 7% immediately.
- **The deeper shift.** Hundreds of millions of people were now going first to ChatGPT, Perplexity or DeepSeek with questions that used to flow automatically to Google Search. Google's mission to "organize the world's information" was intact; its ability to be paid for doing so was not.
- **The irony.** Google was the putative leader in AI and arguably invented the field twice over — the 2012 ImageNet paper (Krizhevsky, Sutskever, Hinton) and the 2017 "Attention Is All You Need" paper that introduced the transformer, written by eight Google Brain scientists. Gemini was competitive with GPT. Yet OpenAI, not Google, brought the technology into public consciousness with ChatGPT in November 2022.

Who is in the fight

- **OpenAI.** ChatGPT reached 100 million monthly users two months after launch — “the fastest-growing consumer application in history” — and 400 million weekly users by February 2025. Backed by Microsoft (an initial \$1 billion in 2019, then a reported \$10 billion). Reportedly lost about \$5 billion in 2024 on high operating costs and unresolved monetisation. Expanding beyond the chatbot: a SoftBank enterprise joint venture, and a \$6.5 billion acquisition of Jony Ive’s hardware startup io.
- **Microsoft.** Bought its way to the front with the OpenAI partnership, then hedged: it built its own cheaper model (Prometheus, for Bing), invested in Mistral and Figure, and licensed Inflection AI’s technology while hiring its founders. Its AI serves the core Microsoft 365 and Azure businesses. Note the pattern — partner, but never depend.
- **Perplexity.** An “answer engine” rather than a search engine, built on OpenAI and Anthropic models accessed by API. Started on subscription, then conceded that “subscriptions alone do not generate enough revenue” and added advertising. It sought CPMs as high as \$60, but advertisers cited limited scale and unproven return. Now pivoting to an agentic browser, Comet, with an eye on Chrome should the US Department of Justice force a divestiture.
- **Meta.** Plays a different game and says so. Zuckerberg’s argument is that Meta does not sell cloud, so commoditising the model layer helps it: “it’s way better if the industry standardizes on the basic tools that we’re using.” LLaMA’s open release made Meta the open-weights leader — though the licence carries a 700-million-monthly-active-user ceiling, which is not what most people mean by open source. In 2025 it began assembling a superintelligence lab with compensation packages reported as high as \$100 million.
- **DeepSeek.** The efficiency shock. In January 2025 the Chinese startup claimed R1 matched frontier performance at a fraction of the cost — one estimate put it at a fifth of the compute of Meta’s LLaMA 3 70B. Nvidia lost \$593 billion of market value in a single day, the largest one-day decline in Wall Street history. Its published techniques — Mixture of Experts (activating only a subset of parameters per request) and distillation (training a small model by querying a large “teacher”) — matter more than its ranking, because they attack cost rather than capability.
- **Anthropic.** Google’s hedge. An initial \$300 million for roughly 10%, later increased by \$2 billion, plus a cloud deal worth over \$3 billion — while Amazon separately committed up to \$4 billion. Google never exercised Microsoft-style influence over it.

What Google has already done

- **Merged its research arms.** Google Brain and DeepMind combined into a single DeepMind in April 2023.
- **Opened part of the stack.** Gemma, a family of open, lightweight models derived from the Gemini research, first released February 2024 (Gemma 3 in March 2025). The context is a leaked internal memo from May 2023 — “We Have No Moat, and Neither Does OpenAI” — which argued that tight control of models is “a fiction” and that the real prize is owning the ecosystem where innovation happens, as Google did with Chrome and Android.
- **Shipped into Search incrementally, not wholesale.** AI Overviews rolled out to the US in May 2024 — an AI summary above a normal results page. Publishers such as the Daily Mail reported clickthrough rates down as much as 44%. After accuracy incidents, Google reduced Overviews from about 15% to 7% of searches by June 2024 — and yet by April 2025 the feature reached over 1.5 billion users monthly, making it the most-used generative AI product in the world. AI Mode, a fully narrative version, arrived as an opt-in tab in May 2025. An “AI Ultra” plan was teased at \$249.99 a month.
- **Read the pattern.** Every move is reversible, rationed, and kept beside the existing product rather than replacing it. That is not indecision — it is what you do when the cost side of the new product is unknown and the old product funds the company.

The unit economics — the heart of the case

- **Revenue per search: ~1.61 cents.** What search advertising earns Google on an average query today.
- **Cost per index query: ~1.060 cents.** What classic Search already costs — crawling, indexing, ranking. Most people assume a search is free to serve. It is not.
- **Added inference cost: ~0.356 cents.** The estimated cost of putting basic generative AI into the answer, holding user behaviour fixed. Inference is the forward pass through the model — the case notes it is roughly proportional to model size (parameters) multiplied by the total length of prompt plus response, and that a single Nvidia GPU draws 700W, more than an average American household.
- **Now do the subtraction.** Today the margin is $1.61 - 1.06$. With basic AI it is $1.61 - (1.06 + 0.356)$. Work out both, then multiply the difference by more than five trillion searches a year. That number is why this is a boardroom problem and not an engineering one. Bring your answer.
- **And note what is NOT in these numbers.** Training cost. Stanford estimated GPT-4 at \$78 million and Gemini at \$191 million to train — against \$900 for Google’s original transformer. Training is a large fixed cost paid once; inference is a small variable cost paid on every request, forever. Confusing the two is the single most common error in AI business cases, and the case is careful to separate them.

Four things that could break the arithmetic

- **Inference costs may fall fast.** Some estimates put the decline at 30% or more per year. Compounded, that changes everything — but it is a forecast, not a fact, and you are being asked to commit before you know.
- **Or user behaviour may inflate them.** Inference scales linearly with the number of words. A user talking to a chatbot writes differently from someone typing keywords: they say please and thank you, ask full questions, supply context, and follow up. A prompt ten times longer costs ten times as much. Sam Altman was asked in April 2025 how much OpenAI had lost to people saying “please” and “thank you”; his answer was “tens of millions of dollars well spent.” It is a joke with a real number inside it.
- **Model choice moves the cost by orders of magnitude.** A larger model, or one that reasons step by step before answering, is dramatically more expensive to serve. This is exactly the Leg A finding arriving on the cost side of the ledger — Med-PaLM 2’s accuracy came from techniques that generate far more tokens per answer.
- **One chat answer may hide several searches.** The case notes that a single generative search may require multiple index queries behind the scenes — the model gathering what it needs to answer. So the 1.060 cents may not be paid once per user question.

If ads do not survive, what replaces them?

- **The advertising problem.** Search advertising works because ads sit beside a page of links, next to a user with visible intent. If the answer arrives as a paragraph of prose, it is unclear where an ad goes or what an advertiser would pay for it. Perplexity’s experience is the live experiment: high asking prices, hesitant advertisers.
- **The subscription problem.** Perplexity and OpenAI both converged on \$20 per month, and conversion to paid is low. The case gives a hard ceiling to reason with: even Microsoft 365, Netflix and Spotify Premium have each topped out around 400 million paid users globally — against Google Search’s 6.25 billion users.
- **And the users who would pay are the expensive ones.** Usage follows a Pareto distribution, which the case makes computable: a share X of users accounts for $X^{0.14}$ of usage. Twenty percent of users generate eighty percent of the usage ($0.20^{0.14} \approx 0.80$). The people most likely to subscribe are the heaviest users — so subscription revenue and inference cost are correlated in the wrong direction. Try the formula on 5% and 50% before class.

The constraints that sit around the numbers

- **Chips.** Nvidia held an estimated 70–95% of the GPU market in 2024; an H100 cost \$25,000 or more, and Meta planned 350,000 of them at up to \$10 billion. Google’s decade-long hedge is its own Tensor Processing Unit — Gemini 2.0 was trained on sixth-generation Trillium TPUs. Owning your own silicon is a cost-structure decision, not a vanity project.
- **Copyright.** Training data is contested ground: the New York Times against OpenAI and Microsoft, and a long list of others (the case tabulates twenty-one suits). Fair use is unresolved. Meanwhile licensing deals are forming — Google–Reddit at a reported \$60 million a year, with OpenAI following. Note that legal exposure is also a cost line.
- **Liability.** Section 230 of the US Communications Decency Act protects platforms from liability for content posted by others — which is why Google is not liable for the pages it links to. If the answer is delivered in Google’s own voice, it is unclear the protection still applies. Hallucination stops being an embarrassment and becomes a liability.
- **Reputation.** Google’s culture had become risk-averse, and the incidents are public — the case reproduces AI Overview insisting the year was 2024 even after being corrected. For a product used by billions, a rare failure is still a very large number of failures.

The decision on the table

- **The narrow question.** Should Google rush to compete directly with the chatbot providers — despite having no working business model for generative AI — or wait, and risk obsolescence? Nothing technical prevents full integration tomorrow. Doing it could jeopardise its most profitable business.
- **The wider question, which the case says matters more.** How could Google reshape the generative AI industry to make it more attractive — for itself and for others? And where in this new value chain should Google position itself to capture the value the technology creates? Note that these are questions about industry structure, not about product features. Come with a view.

READING FOCUS QUESTIONS — bring your answers to class

There are no wrong answers here — these surface how you think, not what you recall. Two of the four are arithmetic; do the arithmetic.

- 1 • Work out Google’s margin per search today, and with basic AI added. Then scale it to five trillion searches a year. What is the annual number, and does it change what you think Google should do?

Clue. The three numbers you need are in the unit-economics section and all are per single search. Do the subtraction twice, then multiply the difference. Two traps to avoid: this is operating margin on the search line, not company profit; and the 0.356 cents assumes user behaviour does not change — which is the assumption most likely to be wrong. Before you conclude, ask yourself which of the four things that could break the arithmetic worries you most.

- 2 • Suppose Google waits five years and inference costs fall 30% per year. Does the problem solve itself?

Clue. Compute it rather than guess: 0.7 raised to the fifth power is what fraction of today's cost? Apply that to the 0.356 cents and redo question 1. Then ask the harder half — what else changes in five years? Prompts get longer, models get more capable and therefore more expensive, and competitors do not stand still. A cost curve falling underneath a demand curve rising is not the same as a problem solving itself.

- 3 • If advertising does not survive the shift to chat answers, what replaces it? Design the business model.

Clue. Take the subscription route seriously and then stress it. At \$20 a month, how many subscribers would Google need to replace search advertising revenue — and how does that compare to the ~400 million ceiling the case says even Netflix and Microsoft 365 hit? Then apply the Pareto formula ($X^{0.14}$): the users who subscribe are the heavy users, so your revenue and your cost grow together. If you do not like subscription, the case leaves other doors open — ads inside answers, enterprise and cloud, licensing, hardware, or making the model layer cheap for everyone and profiting somewhere else entirely. Pick one and defend it.

- 4 • Meta open-sourced LLaMA; Google published the transformer and then released Gemma; DeepSeek published its efficiency techniques. Why would a company give away the thing everyone says is its moat?

Clue. Start from Zuckerberg's own reasoning — Meta does not sell cloud, so commoditising the model layer costs it little and hurts rivals who do sell it. Then read the leaked Google memo again: if control is "a fiction" anyway, the choice is between owning the ecosystem and watching someone else own it. Ask where the profit actually sits if models become interchangeable — distribution, data, chips, or the application on top. And note the cost to Google of publishing "Attention Is All You Need": everything that followed was built on it, by everyone.

- 5 • Connect the two legs. Med-PaLM 2 bought accuracy by generating many times more tokens per answer. What does the AI Wars arithmetic say about where that trade is worth making?

Clue. Put the two side by side. In a hospital, one wrong answer can be catastrophic and a consultation is worth far more than a third of a cent — so an expensive technique is easy to justify. In consumer search, the revenue per event is 1.61 cents and the volume is five trillion. The variable that decides it is not accuracy and not cost in isolation, but the ratio of value-per-answer to cost-per-answer — and, second, the cost of being wrong. Now apply that test to the extraction prompt from Capsule 2 in your own industry.

APPENDIX — THE ECONOMICS, EXPLAINED

This case is written for readers who already have the vocabulary of business economics. Roughly two-thirds of this class does not, and some of these terms carry technical meaning too. Each entry gives the plain-language idea first, then the precise statement — and, where the case supplies one, the number to attach to it. Read this before the case, not after.

The one distinction that matters most. Training cost is a large amount of money spent once to create a model. Inference cost is a small amount of money spent every single time anyone uses it. A business can absorb almost any training cost if enough people use the result; a business can be destroyed by a tiny inference cost if enough people use the result. Everything in this case turns on that asymmetry.

1 • Unit economics

- **Plain.** What one transaction earns and what one transaction costs. Not the company's total profit — the profit on a single unit, in this case a single search. It answers the question "do we make money each time someone uses this, or lose it?"

Technically. Contribution margin per unit = revenue per unit – variable cost per unit. It excludes fixed costs (buildings, salaries, model training), which do not change with volume. It is the right lens whenever volume is enormous, because a per-unit loss multiplied by five trillion is unsurvivable no matter how much cash you hold. Here: 1.61¢ revenue, 1.060¢ existing variable cost, 0.356¢ additional variable cost if generative AI is added.

2 • Fixed cost, variable cost, and capex

- **Plain.** Fixed costs do not move when one more customer arrives; variable costs do. Capital expenditure (capex) is money spent up front on things you will use for years — data centres, chips. In AI, training a model behaves like capex; running it behaves like a variable cost.

Technically. Training GPT-4 was estimated at \$78 million and Gemini at \$191 million (Google's original transformer: \$900). Those are one-time. Inference at 0.356¢ per search across 5 trillion searches is a recurring operating cost that scales with usage and never ends. Note that a company can be simultaneously very profitable and unable to afford a new variable cost — which is precisely Google's position with nearly \$100 billion in cash.

3 • The inference cost formula

- **Plain.** What you pay to run a model once. It goes up with how big the model is and with how much text goes in and comes out. Longer question, longer answer, bigger model — bigger bill, and the relationship is roughly proportional, not gentle.

Technically. The case states inference cost as roughly proportional to model size (parameters) × total length of prompt and response. A forward pass through a trillion-parameter network can require trillions of operations for a query a conventional computer would answer in one. This is the same formula you priced in Capsule 3 as (input tokens × input price + output tokens × output price) × calls × samples. Two consequences the case draws out: a 10× longer prompt is a 10× cost, and step-by-step reasoning is expensive precisely because it generates many more output tokens.

4 • CPM — how advertising is priced

- **Plain.** Cost per thousand impressions: what an advertiser pays to show an ad a thousand times. It is the unit of currency for ad-supported businesses.

Technically. From the Latin mille. Perplexity sought text-ad CPMs as high as \$60 — above a Netflix screen takeover — and advertisers balked on scale, demonstrated return, brand safety and CPM efficiency. The relevance to Google: search advertising works because intent is visible and an ad sits beside a link. Remove the page of links and both the placement and the pricing become unclear.

5 • The Pareto usage distribution

- **Plain.** A small share of users does most of the using. The heaviest 20% typically account for around 80% of activity. It matters here because those same heavy users are the ones who would pay for a subscription — and the ones who cost the most to serve.

Technically. The case gives a computable form: a share X of users represents $X^{0.14}$ of usage, for X between 0 and 1. Check it: $0.20^{0.14} \approx 0.80$. Now try 0.05 and 0.50. The strategic implication is adverse selection — a flat \$20-per-month price attracts exactly the users whose inference costs are highest, so revenue and cost are positively correlated within your paying base. Flat pricing on a usage-priced input is a structural problem, not a pricing detail.

6 • Open weights, closed weights, and the “no moat” argument

- **Plain.** A closed model is one only its owner can run. An open model is published so anyone can download, inspect, modify and run it. Giving away the model sounds like giving away the business — the argument for doing it is that the model is not where the money is.

Technically. Note that “open source” is used loosely here: Meta’s LLaMA 2 licence required a separate grant from Meta for any product above 700 million monthly active users, which is a commercial restriction rather than open source in the conventional sense — “open weights” is the more accurate term. The strategic logic is complementary goods: Meta does not sell cloud, so commoditising models raises the value of what it does sell. The leaked May 2023 Google memo pushes further — control is “a fiction” because anyone can pick a freely available model, so the durable asset is owning the ecosystem where innovation happens, as Chrome and Android did. Google’s Gemma is the answer to that memo.

7 • Mixture of Experts and distillation — attacking cost, not capability

- **Plain.** Two ways to make a model cheaper rather than smarter. Mixture of Experts wakes up only the part of the model relevant to your question instead of all of it. Distillation trains a small cheap model by having it learn from a big expensive one — a student copying a teacher.

Technically. MoE routes each input to a small subset of the parameters, so a model can be very large in total while activating only a fraction per request — cutting inference cost without shrinking capacity. Distillation trains a smaller model on the outputs of a larger one; after DeepSeek demonstrated it, researchers at Berkeley, Stanford and the University of Washington built reasoning models for hundreds of dollars in under a day. Why this matters strategically: both techniques move cost down rather than capability up, and both are published. If cost falls for everyone, no one gains an advantage from it — but the business-model question this case poses gets easier for everyone at once. Watch for that asymmetry.

8 • Section 230 and fair use — the legal cost lines

- **Plain.** Two American legal doctrines that quietly decide how much risk an AI product carries. One protects you from being sued for what other people wrote; the other governs whether you were allowed to train on their writing in the first place.

Technically. Section 230 of the Communications Decency Act of 1996 grants platforms immunity for content posted by users — the reason Google is not liable for the pages it surfaces. A generative answer is delivered in Google’s own voice, so it may not be third-party content at all, and the protection may not hold. Fair use permits limited use of copyrighted material without permission; whether training a commercial model qualifies is unresolved, with over twenty suits pending. Practical reading: hallucination and training-data provenance are not just ethics topics, they are unpriced liabilities on the cost side of this case.

9 • GPU, TPU, and why Google built its own

- **Plain.** The specialised chips that AI runs on. Nvidia sells the industry standard; Google designs its own so it does not have to buy them.

Technically. Graphics processing units perform many similar calculations in parallel, which is what neural networks need. Nvidia held an estimated 70–95% share in 2024, an H100 cost \$25,000 or more, and a single unit draws 700W. Google's Tensor Processing Unit is a decade-old vertical-integration bet; Gemini 2.0 was trained on sixth-generation Trillium TPUs. Read it as a margin decision: at five trillion searches, whoever controls the cost of a forward pass controls the unit economics — which is also why DeepSeek's efficiency claim erased \$593 billion of Nvidia's market value in one day.

How this connects to your own build

The extraction prompt you tune in Capsule 2 has the same shape as this case, four orders of magnitude smaller. It has a revenue per use (what the task is worth to the business), a variable cost per use (tokens), and a set of accuracy techniques that raise both. Before class, put rough numbers on your own example: what is one correct extraction worth, and what would you pay per attempt? If you cannot answer the first, that is the finding — and it is more common than you would think.

SOURCE LINKS

Case (HBS 725-484, purchased — your copy is in the class folder) · store.hbr.org/product/ai-wars-in-2025/725484

Background — the paper that started it · ["Attention Is All You Need" \(2017\), arxiv.org/abs/1706.03762](https://arxiv.org/abs/1706.03762)

Background — the value chain this case sits in · [Wu & Higgins, "Generative AI Value Chain," HBS 724-355](#)

CAVEATS

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2. All figures are estimates reported in the case as of June 2025, several of them from third-party analysts rather than company disclosure. Treat them as order-of-magnitude inputs for reasoning, not as audited accounts — and say so if you quote them.
3. This summary was generated and fact-checked using Anthropic models against the case document. It paraphrases; it does not reproduce the case. Flag anything that looks wrong.